PARTNERSHIP MEMO

Korea–Australia Strategic Energy & Industrial Partnership

Where deeper industrial integration creates strategic value, through which instruments, and where to stay out

Generated from Allied Industrial Base Explorer data and a curated external evidence layer (accessed 2026-11-06); evidence, sources and caveats in the appendix. Data as of 30 October 2026.

Executive Decision Summary

BUILD TOGETHER

Permanent Magnets & Drive Systems: Electric Vehicles components (Korea->Australia). Wind & Hydropower Equipment: Wind towers (Korea->Australia). Battery Materials & Cells: Batteries assembly (Korea->Australia). Grid Equipment & Conductors: Power cables (Korea->Australia). Electric Vehicles: Electric Vehicles assembly (Korea->Australia).

TRADE AND STANDARDS

Common anti-circumvention rules for solar refining (copper wire & rod, polysilicon) and batteries refining (battery materials (foil, electrolyte), cathode materials & precursors, cobalt intermediates & chemicals) and electric vehicles refining (rare-earth metals, rare-earth oxides & compounds). Bind market access for batteries assembly (li-ion cells & packs, other accumulators), batteries components (accumulator parts & separators), electric vehicles components (accumulator parts & separators, electric motors, ferrite & other magnets), wind towers, wind assembly (electricity meters, overhead conductors, power cables), power cables. Keep nuclear special materials; battery recycling feed; semiconductor manufacturing equipment open to global sourcing.

INSTITUTIONS

Upgrade EFA-KEXIM co-financing channel for critical minerals: sector strategy, reciprocal commitments, a KPI dashboard and an annual review. No new MOU.

STAY OUT

Do not create a new bilateral energy MOU or a free-standing MEA institution. Do not put new public money behind Batteries extraction (cobalt ores, lithium minerals, natural graphite) investment that is already committed (evidence KRAU-14. Do not put new public money behind Batteries refining (battery materials (foil, electrolyte), cathode materials & precursors, cobalt intermediates & chemicals) investment that is already committed (evidence KRAU-11.

Why this partnership matters.

- Of 142 technology x stage cells and electro-industrial nodes assessed for the pair, 25 show an active complementarity (a supplier meeting the other's need, a shared chokepoint, a capacity-constrained trusted supplier or a chain specialisation), 0 direct competition and 90 no strategic case. The relationship is not one of general similarity: most of the electro-industrial base needs no bilateral intervention.[1]
- AIB strategic roles: Korea Specialized Technology & Manufacturing Anchor (primary), Scale Manufacturing & Integration Platform (primary), Coordination / Bridge Hub (secondary), Capital & Investment Anchor (secondary); Australia Resource & Processing Anchor (primary).[1]

The central bilateral bargain.

- *Inference.* Australia supplies rare-earth metals, lithium minerals where Korea is short; Korea supplies wind towers, power cables, cathode materials & precursors, accumulator parts & separators where Australia is short; they share no node-level chokepoint where neither supplies.
- *Inference.* From the AIB roles, Korea brings specialised technology and manufacturing, scale manufacturing, coordination across allied chains; Australia brings resources and processing. Public finance runs through KEXIM, K-SURE, KOMIR, KDB on the Korea side and EFA, NRFC, CEFC, ARENA on the Australia side. The trade is market access and demand certainty for trusted capacity.

The 5 highest-priority joint bets (tier 1 by the rules in Section 3, one per compact; ranked by urgency then importance):

PRIORITY	SECTOR	NODES IN THE BET (DIRECTION)	STRATEGIC PROBLEM	KOREA CONTRIBUTION	AUSTRALIA CONTRIBUTION	MAIN INSTRUMENT (LEAD NODE)	TIMING
1	Permanent Magnets & Drive Systems	Electric Vehicles components (Korea->Australia)	strategic manufacturing gap	Electric Vehicles components: trusted capacity (top-quarter position); Wind components: its demand for pooled offtake, and finance for non-Chinese supply	Electric Vehicles components: demand and long-term offtake; Wind components: its demand for pooled offtake, and finance for non-Chinese supply	market access	12-24 months
2	Wind & Hydropower Equipment	Wind towers (Korea->Australia)	strategic manufacturing gap	Wind towers: trusted capacity (12% of world supply)	Wind towers: demand and long-term offtake	market access	12-24 months
3	Battery Materials & Cells	Batteries assembly (Korea->Australia)	geopolitical chokepoint	Batteries assembly: trusted capacity (top-quarter position); Batteries components: trusted capacity (top-quarter position)	Batteries assembly: demand and long-term offtake; Batteries components: demand and long-term offtake	market access	12-24 months
4	Grid Equipment & Conductors	Power cables (Korea->Australia)	strategic manufacturing gap	Power cables: trusted capacity (8% of world supply); Solar refining: its demand for pooled offtake, and finance for non-Chinese supply	Power cables: demand and long-term offtake; Solar refining: its demand for pooled offtake, and finance for non-Chinese supply	market access	12-24 months
5	Electric Vehicles	Electric Vehicles assembly (Korea->Australia)	strategic manufacturing gap	Electric Vehicles assembly: trusted capacity (top-quarter position)	Electric Vehicles assembly: demand and long-term offtake	strategic imports	12-24 months

Sources: 1.

- *Inference.* Outside the electro-industrial compacts, the rules also find fuel-trade lanes (coal refining); they are handled in the trade agenda, not as compacts.

Do not prioritise (Section 10):

- *Recommendation.* Do not create a new bilateral energy MOU or a free-standing MEA institution: KRAU-M3 already exists (finance and projects but no published list); dock the compacts in it.
- *Recommendation.* Do not put new public money behind Batteries extraction (cobalt ores, lithium minerals, natural graphite) investment that is already committed (evidence KRAU-14; KRAU-19; KRAU-20); support is additional only beyond it.

- *Recommendation.* Do not put new public money behind Batteries refining (battery materials (foil, electrolyte), cathode materials & precursors, cobalt intermediates & chemicals) investment that is already committed (evidence KRAU-11; KRAU-12; KRAU-13; KRAU-21); support is additional only beyond it.

Existing mechanisms and the MEA question (Section 8):

- *Inference.* MEA assessment: **upgrade an existing framework (KRAU-M3)**. Gaps: no cross-sector KPI dashboard or review cycle; KRAU-M3 has no published project pipeline.
- *Recommendation.* Do not sign a new energy MOU. Give KRAU-M3 a sector strategy, reciprocal commitments and a KPI dashboard covering the recommended compacts, reviewed every 12 months.

1. Current State of the Relationship

This section sets out what already exists before anything new is proposed. Facts are from the curated evidence layer (accessed 2026-11-06); items marked partial or unverified rest on secondary sources and should be confirmed before citation.

Alliance and plurilateral architecture

- Australia's memberships in OSI's economic-security agreements table: IPEF, MSP, U.S. FTA.[1]
- DFAT notes RCEP entered into force for Korea on 1 Feb 2022 and may be used by exporters as an alternative to KAFTA; the member list shown here is partial.[2]
- Sixth 2+2 met 1 May 2024 with a joint statement; no 2025-26 meeting was found, and the biennial cadence implies the next falls around 2026.[3]
- Lee noted large Korean investment in Australian lithium, critical-minerals and rare-earth projects; no joint statement or deliverable found.[4]
- ASPI lists the MSP, which Korea chairs in 2025, among the frameworks for Australia-Korea minerals cooperation; Australian membership is not stated in that article.[5]
- Korea chaired the Minerals Security Partnership until June 2026; a February 2026 report says its 17 members carried into the successor FORGE; Korean public lenders joined the MSP Finance Network; no signed US-Korea critical-minerals MOU was found.[6]
- On 24 Jun 2026 Korea Zinc's chair met Albanese in Canberra on supply chains and smelting competitiveness and agreed to explore cooperation; Project Crucible is the US$7.4bn Tennessee smelter, not an Australian investment.[7]
- Albanese called Korea a trusted partner in critical minerals and clean energy at a Sep 2026 UNGA meeting, the third bilateral after the G7 and APEC; no deliverable reported; day approximate.[8]

Trade architecture and frictions

- In the OSI energy basket, Korea imported US$18.1bn from Australia and Australia imported US$10.9bn from Korea (bilateral flows 2024, summed over the technology x stage cells).[1]
- Signed in Seoul 8 Apr 2014 and in force 12 Dec 2014; Australia reached duty-free on all Korean imports from 1 Jan 2021 and Korean tariffs on Australian goods were phased out by 2023; RCEP is an alternative route.[9]

Industrial and energy agreements

- The 2021 partnership commits both to critical-mineral supply-chain resilience and a ministers' MoU, with a Critical Minerals Working Group on supply-chain mapping, R&D and investment.[10]
- Revised 1988 MOU signed 30 Oct 2025 at APEC Gyeongju names critical minerals and hydrogen, joint third-market entry and resumed biennial consultations in Australia from 2026.[11]

Investment vehicle, project pipeline and financing

- Australia announced an initial A$50m for the critical-minerals partnership, with Korea to match subject to domestic processes; USD value approximate.[12]
- EFA and the Export-Import Bank of Korea signed an MoU in Sep 2023 on co-financing in critical minerals and related sectors; no amounts given, and the day is approximate.[13]

Corporate relationships in the recommended compacts

- Binding term sheet Jan 2022 and full offtake May 2022 for spodumene concentrate; reported buyer termination right if production had not begun by 1 Sep 2025.[14]
- Oct 2022 five-year offtake (option for five more) with SK On taking a 10% stake in the ASX-listed Lake Resources.[15]
- Seven-year NdPr oxide offtake (600 t/yr rising to 1,500 t/yr) with conditions to be met by end-Jun 2026; KEXIM issued a letter of intent for up to US$150m debt.[16]
- Two Korean government-owned banks funded US$460m for the Gwangyang plant, non-recourse to Pilbara Minerals; date approximate.[17]
- Feb 2024 second agreement for up to 85,000 t of concentrate from Mt Holland, after a 2022 deal for 50,000 t of hydroxide; no value disclosed.[18]
- Nov 2024 agreement lets SK On take up to 15,000 t of Korea-made hydroxide from Australian-sourced spodumene; POSCO Future M is to take 20,000 t a year; day approximate.[19]
- Plant converts Pilgangoora spodumene to battery-grade hydroxide (43,000 t/yr nameplate); train 1 ramped in 2024 and train 2 was completed in early 2025; no 2026 production data found; day approximate.[20]
- ASM's Korean plant opened in 2022 and produces neodymium metals and alloys (designed 5,200 t/yr); its Dubbo (NSW) mine, intended to feed it, has unclear funding in the sources seen.[21]
- POSCO's 4 Sep 2026 release says it acquired a stake in a high-quality lithium mine owned by Mineral Resources in 2026; mine name, stake size and amount are not given.[22]
- Final investment decision in May 2026 after an Australian offtake commitment under the Strategic Reserve; construction began 8 Oct 2026; credit agencies from the US, Canada, Korea and Germany support it; the article's A$1.2bn figure is ambiguous.[23]
- Unverified items are left out of this section; they stay in the evidence register (evidence_register.csv) and the evidence layer.[1]
- *Inference.* Working channels today: Comprehensive Strategic Partnership and Critical Minerals Working Group (finance); EFA-KEXIM co-financing channel for critical minerals (a project pipeline and finance). What the relationship lacks: no cross-sector KPI dashboard or review cycle; KRAU-M3 has no published project pipeline.

2. What Each Country Is Good At — and What It Needs

Cell codes: position / need for **Korea · Australia**. Position S strong (top quarter of contenders and above the capability floor), M contender, W not a contender; need H high (supply-chain need >= 0.67, or deployment need >= 0.60 at Deployment), L low; - not measured. Source: strategic_position_tbl (engine choices are in country_profiles.csv).

TECHNOLOGY	EXTRACTION	REFINING	COMPONENTS	ASSEMBLY	DEPLOYMENT	RECOVERY
Batteries	W/H · S/L	W/H · M/H	M/H · W/H	S/H · W/H	S/L · S/H	-/H · -/H
Coal	W/H · S/L	W/H · M/H	S/L · W/L	S/L · W/L	S/L · M/L	-/- · -/-
Electric Grid	W/H · W/H	W/H · W/L	M/H · W/L	M/L · W/L	-/L · -/H	-/- · -/-
Electric Vehicles	W/H · S/H	W/H · W/H	S/H · W/H	S/L · W/H	S/H · S/L	-/- · -/-
Gas	W/H · S/L	W/H · M/H	M/L · W/L	S/L · W/L	S/L · M/L	-/- · -/-

TECHNOLOGY	EXTRACTION	REFINING	COMPONENTS	ASSEMBLY	DEPLOYMENT	RECOVERY
Geothermal	-/- · -/-	-/- · -/-	-/- · -/-	-/- · -/-	M/H · S/L	-/- · -/-
Green Hydrogen	W/- · W/-	W/- · W/-	-/- · -/-	-/- · -/-	-/H · -/L	-/- · -/-
Nuclear	W/H · M/L	W/H · M/L	S/L · W/L	S/L · W/L	S/H · M/L	-/- · -/-
Oil	W/H · M/H	M/L · W/L	M/H · M/H	S/L · W/L	S/L · S/H	-/L · -/H
Solar	W/H · W/H	W/H · W/H	M/L · W/L	M/L · W/H	M/H · S/H	-/- · -/-
Wind	W/- · S/-	W/H · W/H	M/H · W/H	M/L · W/H	S/H · S/H	-/- · -/-

Sources: 1.

Electro-industrial nodes where either country has a position or a need (AIB division of labour, 2024 trade; share = world export or production share; need is the node's technology x stage need, inherited, so a proxy; demand = share of allied demand). 61 of 76 nodes:

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

Table A1 in the appendix (61 rows) carries this table in full; the CSV outputs carry every column.

- Revealed policy priority (NIPO / IEA PAMS within-country percentile; policy never enters the classification) is highest for Korea in: Batteries assembly (0.96); Green Hydrogen assembly (0.95); Nuclear assembly (0.92); Nuclear components (0.90); Electric Vehicles recovery (0.87); for Australia in: Green Hydrogen assembly (0.99); Green Hydrogen components (0.96); Solar assembly (0.92); Electric Vehicles extraction (0.90); Batteries refining (0.89).[1]
- Evidence quality: 40 of 66 cells have high stage-specific evidence for both countries; 19 units are flagged low-evidence.[1]

3. Bilateral Complementarity Map

State counts over all units: a_supplies_b_needs 7; b_supplies_a_needs 3; complementary_specialization 7; shared_vulnerability 6; capacity_constrained_trusted_supply 2; low_strategic_case 90; deployment_demand 9; not_assessed 18.[1]

Inference. Rules (Section 3 of docs/partnership_memos.md): two strong positions are **direct competition**, never complementarity; a supplier must be strong (or a contender the engine names as the other's trusted partner) and the buyer must have high need, or material demand where trusted capacity is short or a chokepoint exists; **capacity-constrained** where the AIB finds allied capacity short of demand; **shared vulnerability** where both need the node, neither supplies it and a coercive supplier holds at least half of world supply.

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

Table A2 in the appendix (25 rows) carries this table in full; the CSV outputs carry every column.

- Low-priority / open-market nodes (69): wafers, polysilicon, synthetic graphite (anode-grade), solar cells, pv modules & generators, li-ion cells & packs, rare-earth oxides & compounds, natural graphite, battery materials (foil, electrolyte), cobalt intermediates & chemicals, lithium chemicals, battery recycling feed, other accumulators, manganese dioxide (emd), nickel intermediates & chemicals, cobalt ores, uranium conversion & enrichment, fuel assemblies & zirconium, reactor parts, nuclear reactors & enrichment machinery, uranium/fuel extraction: aggregated low-coverage sub-sectors (spent fuel & nuclear scrap, uranium ores), electric motors, rare-earth permanent magnets, electric & plug-in cars, electric two-wheelers & works trucks, electric buses, trucks & tractors, rare-earth ores (monazite), ferrite & other magnets, motor & generator parts, wind turbine generating sets, pv cover glass, wind blades & hubs, copper ores & concentrates, copper smelting & refining, electricity meters, overhead conductors, automatic controllers, static converters & inverters, switchgear & protection, transformers, copper wire & rod, grain-oriented electrical steel (goes), insulators & bushings, power

electronics components, ac generators, piezo-electric quartz, refrigerants, heat pumps & ac units, solar thermal water heaters, gallium, germanium & indium, electrolysers, refrigeration & heat-pump compressors, power semiconductors, sensors & controller parts, electric furnaces & heating, furnace parts, integrated circuits, hydro turbine parts, graphite electrodes, blade core wood (balsa), semiconductor manufacturing equipment, hydrogen, ammonia & methanol, nuclear special materials, silver powder, electric rail vehicles, direct-reduced iron (dri/hbi), aluminium wire rod, hydro turbines, pgms (iridium, ruthenium).[1]

- Deployment cells are demand context, not supply relations: high deployment need in Green Hydrogen (KOR); Batteries (AUS); Nuclear (KOR); Electric Vehicles (KOR); Solar (KOR and AUS); Wind (KOR and AUS); Electric Grid (AUS); Geothermal (KOR); Oil (AUS).[1]

4. Trade Agenda

- *Inference.* **What Korea should export to Australia:** Batteries assembly (li-ion cells & packs, other accumulators); Batteries components (accumulator parts & separators); Electric Vehicles components (accumulator parts & separators, electric motors, ferrite & other magnets); Electric Vehicles assembly (electric & plug-in cars, electric buses, trucks & tractors, electric two-wheelers & works trucks); Wind towers; Wind assembly (electricity meters, overhead conductors, power cables); Power cables; Cathode materials & precursors; Accumulator parts & separators; Non-oriented electrical steel (NGO); Gas assembly; Coal components; Coal assembly.
- *Inference.* **What Australia should export to Korea:** Rare-earth metals; Batteries extraction (cobalt ores, lithium minerals, natural graphite); Lithium minerals; Electric Vehicles extraction (cobalt ores, lithium minerals, natural graphite); Gas extraction; Coal extraction.
- *Inference.* **Where tariffs would undermine allied specialisation:** Batteries|Assembly, Batteries|Components, Electric Vehicles|Components, wind-towers, Wind|Assembly, power-cables.
- *Inference.* **Procurement, standards and content rules to recognise mutually:** .
- *Inference.* **Where third-country circumvention safeguards are genuinely required:** Solar|Refining, Batteries|Refining, Electric Vehicles|Refining; elsewhere they would add cost without security.
- *Inference.* **Where unrestricted global sourcing is acceptable:** nuclear-special-materials, battery-recycling-feed, semiconductor-manufacturing-equipment, copper-ores-and-concentrates, silver-powder, uranium-conversion-and-enrichment.

Table A3 in the appendix (18 rows) carries this table in full; the CSV outputs carry every column.

5. Investment Agenda

Korea → Australia investment (Korea capital and firms expanding in Australia)

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

ID	NODE	CAPITAL / CAPABILITY OWNER	DESTINATION	DEMAND CASE	PRIVATE ACTORS (EVIDENCE)	PUBLIC INSTRUMENT	ADDITIONALITY	HORIZON
I02	Batteries assembly (li-ion cells & packs, other accumulators)	KOR	AUS	supply-chain need 0.83 (KOR) / 0.85 (AUS)		market access	high	2-3 years
I03	Batteries components (accumulator parts & separators)	KOR	AUS	supply-chain need 0.73 (KOR) / 0.72 (AUS)		market access	medium	2-3 years

ID	NODE	CAPITAL / CAPABILITY OWNER	DESTINATION	DEMAND CASE	PRIVATE ACTORS (EVIDENCE)	PUBLIC INSTRUMENT	ADDITIONALITY	HORIZON
I04	Electric Vehicles components (accumulator parts & separators, electric motors, ferrite & other magnets)	KOR	AUS	supply-chain need 0.97 (KOR) / 0.76 (AUS)		market access	medium	2-3 years
I05	Electric Vehicles assembly (electric & plug-in cars, electric buses, trucks & tractors, electric two-wheelers & works trucks)	KOR	AUS	supply-chain need 0.37 (KOR) / 0.70 (AUS)		export credit	medium	2-3 years
I06	Wind towers	KOR	AUS	AUS takes 5% of allied demand (5% of allied net imports); AIB class LOW INTERVENTION		market access	medium	2-3 years
I07	Wind assembly (electricity meters, overhead conductors, power cables)	KOR	AUS	supply-chain need 0.64 (KOR) / 0.83 (AUS)		market access	medium	2-3 years
I08	Power cables	KOR	AUS	AUS takes 4% of allied demand (4% of allied net imports); AIB class LOW INTERVENTION		market access	medium	2-3 years
I09	Cathode materials & precursors	KOR	AUS	AUS takes 0% of allied demand (0% of allied net imports); AIB class BUILD		export credit	high	2-3 years
I10	Accumulator parts & separators	KOR	AUS	AUS takes 0% of allied demand (0% of allied net imports); AIB class BUILD		standards alignment	high	2-3 years
I11	Non-oriented electrical steel (NGO)	KOR	AUS	AUS takes 0% of allied demand (0% of allied net imports); AIB class BUILD		standards alignment	high	2-3 years

Sources: 1.

Australia → Korea investment (Australia capital and firms expanding in Korea)

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

ID	NODE	CAPITAL / CAPABILITY OWNER	DESTINATION	DEMAND CASE	PRIVATE ACTORS (EVIDENCE)	PUBLIC INSTRUMENT	ADDITIONALITY	HORIZON
I01	Rare-earth metals	AUS	KOR	KOR takes 2% of allied demand (2% of allied net imports); AIB class BUILD	KSM Metals	export credit	low	2-3 years

Sources: 1.

Joint investment (at home or in third countries)

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

ID	NODE	CAPITAL / CAPABILITY OWNER	DESTINATION	DEMAND CASE	PRIVATE ACTORS (EVIDENCE)	PUBLIC INSTRUMENT	ADDITIONALITY	HORIZON
I12	Batteries extraction (cobalt ores, lithium minerals, natural graphite)	KOR+AUS	third countries (CHL; ARG; BRA) and either partner	supply-chain need 1.00 (KOR) / 0.12 (AUS)	POSCO Holdings; Mineral Resources; LG Energy Solution	anchor offtake	low	2-3 years
I13	Electric Vehicles extraction (cobalt ores, lithium minerals, natural graphite)	KOR+AUS	third countries (CHL; ARG; BRA) and either partner	supply-chain need 1.00 (KOR) / 0.72 (AUS)	Arafura Rare Earths; POSCO Holdings; Mineral Resources	anchor offtake	low	2-3 years

Sources: 1.

- *Inference.* Additionality = mean of three named components: the market failure (chokepoint 1, capacity constraint 0.75, gap or emerging bottleneck 0.5, none 0), one minus an existing commitment in the evidence layer (committed 1, announced 0.5) and one minus a working market (the buyer already takes 30% or more of its imports from the supplier). It is **low** whenever there is no failure or the investment is already committed: public support would then not be additional.

6. Third-Country and Development Cooperation

- From the development-partnership pairs (scripts/93: a donor's need meets a developing economy's credible Catch Up pathway in the same step), Korea and Australia are both donors at 10 recipient steps. Ranked: both donors' own need first, then steps inside a recommended compact.[1]
- *Inference.* Roles: Korea brings specialised technology and components and equipment and manufacturing at scale; Australia brings resources and processing (AIB strategic roles).

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ID	RECIPIENT	TECHNOLOGY	STAGE	COMPACT	RECIPIENT CHOICE	RECIPIENT ANCHOR	KOREA INTEREST	AUSTRALIA INTEREST
X01	Indonesia	Batteries	Refining	SC-BATTERIES	preserve	direct supply share; direct export share; operating capacity; committed pipeline	own catch up need (KOR)	own catch up need (AUS)
X02	Argentina	Batteries	Refining	SC-BATTERIES	catch up	direct supply share; operating capacity; committed pipeline	own catch up need (KOR)	own catch up need (AUS)
X03	Democratic Republic of Congo	Batteries	Refining	SC-BATTERIES	catch up	direct export share; operating capacity	own catch up need (KOR)	own catch up need (AUS)
X04	Malaysia	Batteries	Refining	SC-BATTERIES	catch up	direct export share	own catch up need (KOR)	own catch up need (AUS)
X05	Madagascar	Batteries	Refining	SC-BATTERIES	catch up	direct export share; operating capacity	own catch up need (KOR)	own catch up need (AUS)
X06	Philippines	Batteries	Refining	SC-BATTERIES	catch up	direct export share; operating capacity	own catch up need (KOR)	own catch up need (AUS)
X07	India	Batteries	Refining	SC-BATTERIES	catch up	direct export share	own catch up need (KOR)	own catch up need (AUS)

ID	RECIPIENT	TECHNOLOGY	STAGE	COMPACT	RECIPIENT CHOICE	RECIPIENT ANCHOR	KOREA INTEREST	AUSTRALIA INTEREST
X08	South Africa	Batteries	Refining	SC-BATTERIES	catch up	direct export share	own catch up need (KOR)	own catch up need (AUS)
X09	Papua New Guinea	Batteries	Refining	SC-BATTERIES	catch up	direct export share	own catch up need (KOR)	own catch up need (AUS)
X10	Thailand	Batteries	Refining	SC-BATTERIES	catch up	direct export share	own catch up need (KOR)	own catch up need (AUS)

Sources: 1.

- DFAT notes RCEP entered into force for Korea on 1 Feb 2022 and may be used by exporters as an alternative to KAFTA; the member list shown here is partial.[2]
- ASPI lists the MSP, which Korea chairs in 2025, among the frameworks for Australia-Korea minerals cooperation; Australian membership is not stated in that article.[5]
- Korea chaired the Minerals Security Partnership until June 2026; a February 2026 report says its 17 members carried into the successor FORGE; Korean public lenders joined the MSP Finance Network; no signed US-Korea critical-minerals MOU was found.[6]
- Final investment decision in May 2026 after an Australian offtake commitment under the Strategic Reserve; construction began 8 Oct 2026; credit agencies from the US, Canada, Korea and Germany support it; the article's A$1.2bn figure is ambiguous.[23]

7. Recommended Sectoral Compacts

Inference. 2 compacts pass (rule: at least one joint bet, at least two active nodes and both countries holding an AIB compact role; or, labelled **evidence-led**, an existing bilateral pipeline or at least 2 bilateral project items where OSI measures a need but cannot see the partner's capability). The hypothesis test for every AIB compact:

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

RANK	COMPACT	VERDICT	JOINT BETS	COORDINATE	KOREA ROLE	AUSTRALIA ROLE	BILATERAL MECHANISMS	RECOMMENDED
1	Battery Materials & Cells	evidence-led candidate	0	3	Specialized technology supplier	Resource supplier	KRAU-M2; KRAU-M3	TRUE
2	Permanent Magnets & Drive Systems	evidence-led candidate	0	2	Specialized technology supplier	Processing anchor	KRAU-M2; KRAU-M3	TRUE
3	Wind & Hydropower Equipment	secondary	1	0	Specialized technology supplier	Bridge / coordination partner		FALSE
4	Grid Equipment & Conductors	secondary	1	0	Scale manufacturer			FALSE
5	Electric Vehicles	not a bilateral priority	0	0	Scale manufacturer	Bridge / coordination partner		FALSE
6	Heat Pumps & Cooling	not a bilateral priority	0	0	Scale manufacturer			FALSE

RANK	COMPACT	VERDICT	JOINT BETS	COORDINATE	KOREA ROLE	AUSTRALIA ROLE	BILATERAL MECHANISMS	RECOMMENDED
7	Industrial Electrification & Hydrogen	not a bilateral priority	0	0	Scale manufacturer			FALSE
8	Nuclear Fuel & Reactor Systems	not a bilateral priority	0	0	Scale manufacturer	Resource supplier		FALSE
9	Semiconductors & Power Electronics	not a bilateral priority	0	0	Specialized technology supplier			FALSE
10	Solar Manufacturing	not a bilateral priority	0	0	Specialized technology supplier			FALSE

Sources: 1.

1. Battery Materials & Cells (evidence-led)

- *Inference.* **Basis: external evidence.** OSI trade data show no bilateral complementarity at this compact's nodes (Australia world share: cathode materials & precursors 0.0%, accumulator parts & separators 0.0%, lithium minerals 52.7%, synthetic graphite (anode-grade) 0.0%, li-ion cells & packs 0.0%, natural graphite 0.0%, battery materials (foil, electrolyte) 0.0%, cobalt intermediates & chemicals 1.6%, lithium chemicals 2.0%, battery recycling feed 0.0%, other accumulators 0.2%, manganese dioxide (emd) 0.0%, nickel intermediates & chemicals 3.5%, cobalt ores 1.3%); the case rests on the bilateral pipeline and corporate evidence listed at the end of this compact.
- Need that OSI does measure in the compact: lithium minerals (Korea high 1.00, 14% of allied demand; Australia low (cell 0.72; not a material buyer)); synthetic graphite (anode-grade) (Korea high 0.99, 6% of allied demand; Australia low (cell 0.83; not a material buyer)); li-ion cells & packs (Korea low (cell 0.97; not a material buyer), 4% of allied demand; Australia high 0.85); natural graphite (Korea high 1.00, 9% of allied demand; Australia low (cell 0.72; not a material buyer)); battery materials (foil, electrolyte) (Korea high 0.99, 16% of allied demand; Australia low (cell 0.83; not a material buyer)); cobalt intermediates & chemicals (Korea high 0.99, 6% of allied demand; Australia low (cell 0.83; not a material buyer)); lithium chemicals (Korea high 0.99, 50% of allied demand; Australia low (cell 0.83; not a material buyer)); battery recycling feed (Korea high 0.94, 20% of allied demand; Australia low (cell 0.95; not a material buyer)); other accumulators (Korea low (cell 0.83; not a material buyer), 1% of allied demand; Australia high 0.85).[1]

Strategic objective

- AIB mission: Build missing capacity in Li-ion cells & packs and lithium chemicals and manganese dioxide (EMD) (refined chemicals); targeted: preserve and second-source specialized production in lithium chemicals (refined chemicals)..[1]
- *Recommendation.* Change the outcome at cathode materials & precursors, accumulator parts & separators, lithium minerals: geopolitical chokepoint; trusted partner capacity constraint; no material intervention.

Why this pair?

- *Inference.* Cathode materials & precursors: Complementary specialisation (KOR is strong here and the other country is strong at another step of the same chain). Accumulator parts & separators: Complementary specialisation (KOR is strong here and the other country is strong at another step of the same chain). Lithium minerals: Australia supplies / Korea needs (AUS is a capable trusted supplier and the other side needs the node).

- AIB compact roles: Korea Specialized technology supplier (lead node cathode-materials-and-precursors); Australia Resource supplier (lead node lithium-minerals).[1]

Supply-chain architecture

STAGE	UNIT	BASIS	KOREA POSITION	AUSTRALIA POSITION	DESIRED BILATERAL ARRANGEMENT	TREATMENT	INSTRUMENT
Extraction	lithium-minerals	AIB node	weak	strong	b supplies a needs	COORDINATE	
Refining	cathode-materials-and-precursors	AIB node	strong	weak	complementary specialization	COORDINATE	standards alignment
Components	accumulator-parts-and-separators	AIB node	strong	weak	complementary specialization	COORDINATE	standards alignment
Assembly	li-ion-cells-and-packs	AIB node	moderate	weak	low strategic case	LOW INTERVENTION	
Deployment	Batteries/Deployment	OSI cell (no AIB node at this stage)	strong	strong	deployment demand	DEMAND CONTEXT	
Recovery	battery-recycling-feed	AIB node	weak	weak	low strategic case	LOW INTERVENTION	

Sources: 1.

Reciprocal bargain

- Korea provides: supply of cathode materials & precursors, accumulator parts & separators; long-term demand and offtake for lithium minerals; procurement recognition of the partner as trusted supply; finance for the compact's nodes (KEXIM, K-SURE, KOMIR, KDB).
- Australia provides: capacity and technology in lithium minerals; long-term demand for cathode materials & precursors, accumulator parts & separators; procurement recognition of the partner as trusted supply; finance for capacity in the partner market (EFA, NRFC, CEFC, ARENA).
- Jointly: common specifications and qualification.
- Third-country partners needed: JPN, CHL, ARG, BRA.
- Better placed than this pair (largest trusted supplier once China is removed, AIB): DEU at synthetic graphite (anode-grade); POL at li-ion cells & packs; MDG at natural graphite; TWN at battery materials (foil, electrolyte); FIN at cobalt intermediates & chemicals; CHL at lithium chemicals; VNM at other accumulators; JPN at manganese dioxide (emd). The bilateral compact should not try to replace them there.[1]

Policy instruments (minimum necessary)

- *Recommendation.* standards_alignment for cathode materials & precursors; standards_alignment for accumulator parts & separators; NA for lithium minerals

Priority transactions

- *Recommendation.* Tariff binding for Korea batteries assembly (li-ion cells & packs, other accumulators) entering Australia (D01).
- *Recommendation.* Long-term offtake by Korea buyers for Australia batteries extraction (cobalt ores, lithium minerals, natural graphite), sized to underwrite new capacity (candidate hosts: CHL, ARG, BRA) (D08).

Success by 2030

- Cathode materials & precursors: coercive share of supply to the two buyers below 50% (CHN now holds 92% of world supply); Korea's share of Australia's imports up from 1%.
- Accumulator parts & separators: allied residual gap under the China shock closed (now 37% of allied demand); Korea's share of Australia's imports up from 1%.
- Lithium minerals: allied residual gap under the China shock closed (now 9% of allied demand); Australia's share of Korea's imports up from 25%.
- The AIB node class moves from BUILD to COORDINATE or LOW INTERVENTION on the next run.

Failure / exit conditions

- No signed offtake or financing within 24 months of launch: drop the node from the compact.
- The coercive share of world supply falls below the chokepoint line, or the AIB class turns LOW INTERVENTION: move to open-market treatment.
- Projects proceed on commercial terms without public support: stop the support (it is not additional).

Evidence (external)

- Binding term sheet Jan 2022 and full offtake May 2022 for spodumene concentrate; reported buyer termination right if production had not begun by 1 Sep 2025.[14]
- Oct 2022 five-year offtake (option for five more) with SK On taking a 10% stake in the ASX-listed Lake Resources.[15]
- Two Korean government-owned banks funded US$460m for the Gwangyang plant, non-recourse to Pilbara Minerals; date approximate.[17]
- Feb 2024 second agreement for up to 85,000 t of concentrate from Mt Holland, after a 2022 deal for 50,000 t of hydroxide; no value disclosed.[18]
- Nov 2024 agreement lets SK On take up to 15,000 t of Korea-made hydroxide from Australian-sourced spodumene; POSCO Future M is to take 20,000 t a year; day approximate.[19]
- Plant converts Pilgangoora spodumene to battery-grade hydroxide (43,000 t/yr nameplate); train 1 ramped in 2024 and train 2 was completed in early 2025; no 2026 production data found; day approximate.[20]
- POSCO's 4 Sep 2026 release says it acquired a stake in a high-quality lithium mine owned by Mineral Resources in 2026; mine name, stake size and amount are not given.[22]

2. Permanent Magnets & Drive Systems (evidence-led)

- *Inference.* **Basis: external evidence.** OSI trade data show no bilateral complementarity at this compact's nodes (Australia world share: rare-earth metals 25.1%, non-oriented electrical steel (ngo) 0.0%, rare-earth oxides & compounds 0.0%, electric motors 0.1%, rare-earth permanent magnets 0.0%, rare-earth ores (monazite) 5.1%, ferrite & other magnets 0.0%, motor & generator parts 0.0%, ac generators 0.1%); the case rests on the bilateral pipeline and corporate evidence listed at the end of this compact.
- Need that OSI does measure in the compact: rare-earth oxides & compounds (Korea high 0.99, 7% of allied demand; Australia low (cell 0.86; not a material buyer)); electric motors (Korea high 0.97, 5% of allied demand; Australia low (cell 0.76; not a material buyer)); rare-earth permanent magnets (Korea high 0.97, 7% of allied demand; Australia low (cell 0.92; not a material buyer)); ac generators (Korea high 0.80, 8% of allied demand; Australia high 0.92).[1]

Strategic objective

- AIB mission: Build missing capacity in rare-earth permanent magnets (magnets & core steel)..[1]
- *Recommendation.* Change the outcome at rare-earth metals, non-oriented electrical steel (ngo): geopolitical chokepoint; trusted partner capacity constraint.

Why this pair?

- *Inference.* Rare-earth metals: Complementary specialisation (AUS is strong here and the other country is strong at another step of the same chain). Non-oriented electrical steel (NGO): Complementary specialisation (KOR is strong here and the other country is strong at another step of the same chain).
- AIB compact roles: Korea Specialized technology supplier (lead node non-oriented-electrical-steel-ngo); Australia Processing anchor (lead node rare-earth-metals).[1]

Supply-chain architecture

STAGE	UNIT	BASIS	KOREA POSITION	AUSTRALIA POSITION	DESIRED BILATERAL ARRANGEMENT	TREATMENT	INSTRUMENT
Extraction	rare-earth-ores-monazite	AIB node	weak	moderate	low strategic case	LOW INTERVENTION	
Refining	rare-earth-metals	AIB node	weak	strong	complementary specialization	COORDINATE	standards alignment
Components	rare-earth-permanent-magnets	AIB node	weak	weak	low strategic case	LOW INTERVENTION	
Assembly	electric-motors	AIB node	moderate	weak	low strategic case	LOW INTERVENTION	
Deployment	Electric Vehicles/Deployment	OSI cell (no AIB node at this stage)	strong	strong	deployment demand	DEMAND CONTEXT	
Recovery	Electric Vehicles/Recovery	OSI cell (no AIB node at this stage)	not measured	not measured	not assessed	INSUFFICIENT DATA	

Sources: 1.

Reciprocal bargain

- Korea provides: supply of non-oriented electrical steel (ngo); long-term demand and offtake for rare-earth metals; procurement recognition of the partner as trusted supply; finance for the compact's nodes (KEXIM, K-SURE, KOMIR, KDB).
- Australia provides: capacity and technology in rare-earth metals; long-term demand for non-oriented electrical steel (ngo); procurement recognition of the partner as trusted supply; finance for capacity in the partner market (EFA, NRFC, CEFC, ARENA).
- Jointly: common specifications and qualification.
- Third-country partners needed: THA, USA, MYS, JPN, TWN.
- Better placed than this pair (largest trusted supplier once China is removed, AIB): MYS at rare-earth oxides & compounds; DEU at electric motors; JPN at rare-earth permanent magnets; DEU at ferrite & other magnets; DEU at motor & generator parts. The bilateral compact should not try to replace them there.[1]

Policy instruments (minimum necessary)

- *Recommendation.* standards_alignment for rare-earth metals; standards_alignment for non-oriented electrical steel (ngo)

Priority transactions

- *Recommendation.* Tariff binding for Korea electric vehicles components (accumulator parts & separators, electric motors, ferrite & other magnets) entering Australia (D03).

- *Recommendation.* Long-term offtake by Korea and Australia buyers for non-coercive wind refining (copper wire & rod, rare-earth metals, rare-earth oxides & compounds), sized to underwrite new capacity (candidate hosts: THA, USA, MYS) (D09).

Success by 2030

- Rare-earth metals: coercive share of supply to the two buyers below 50% (CHN now holds 84% of world supply); Australia's share of Korea's imports up from 0%.
- Non-oriented electrical steel (NGO): allied residual gap under the China shock closed (now 17% of allied demand); Korea's share of Australia's imports up from 0%.
- The AIB node class moves from BUILD to COORDINATE or LOW INTERVENTION on the next run.

Failure / exit conditions

- No signed offtake or financing within 24 months of launch: drop the node from the compact.
- The coercive share of world supply falls below the chokepoint line, or the AIB class turns LOW INTERVENTION: move to open-market treatment.
- Projects proceed on commercial terms without public support: stop the support (it is not additional).

Evidence (external)

- Seven-year NdPr oxide offtake (600 t/yr rising to 1,500 t/yr) with conditions to be met by end-Jun 2026; KEXIM issued a letter of intent for up to US$150m debt.[16]
- ASM's Korean plant opened in 2022 and produces neodymium metals and alloys (designed 5,200 t/yr); its Dubbo (NSW) mine, intended to feed it, has unclear funding in the sources seen.[21]
- ASM made its first commercial sale of terbium and dysprosium from its Korean plant in Jul 2025, to a Neo Performance Materials subsidiary; the buyer is not a Korean firm per the source; day approximate.[24]
- Final investment decision in May 2026 after an Australian offtake commitment under the Strategic Reserve; construction began 8 Oct 2026; credit agencies from the US, Canada, Korea and Germany support it; the article's A$1.2bn figure is ambiguous.[23]

8. Master Energy Agreement Assessment

- *Inference.* Rule: **existing** if every recommended compact is covered by a mechanism with both a pipeline and finance and a cross-sector mechanism has KPIs and a review cycle; **upgrade** if at least 50% of compacts are covered by a mechanism with finance or a pipeline, or a cross-sector mechanism with a pipeline exists; **new MEA** otherwise.
- *Inference.* Result: **upgrade**; coverage 100% of recommended compacts; cross-sector mechanism with KPIs and review: no.

COMPACT	ANY MECHANISM	FINANCE OR PIPELINE	PIPELINE AND FINANCE
SC-BATTERIES	TRUE	TRUE	FALSE
SC-MAGNETS	TRUE	TRUE	FALSE

Sources: 1.

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

ID	MECHANISM	STATUS	SCOPE	PIPELINE	FINANCE	KPIS	REVIEW	RECIPROCAL
KRAU-M1	KAFTA and RCEP	in force	cross sector	FALSE	FALSE	FALSE	partial	TRUE

ID	MECHANISM	STATUS	SCOPE	PIPELINE	FINANCE	KPIS	REVIEW	RECIPROCAL
KRAU-M2	Comprehensive Strategic Partnership and Critical Minerals Working Group	in force	SC-BATTERIES/SC-MAGNETS/cross sector	FALSE	partial	FALSE	partial	TRUE
KRAU-M3	EFA-KEXIM co-financing channel for critical minerals	signed	SC-MAGNETS/SC-BATTERIES	partial	TRUE	FALSE	FALSE	TRUE

Sources: 1.

Recommended: upgrade EFA-KEXIM co-financing channel for critical minerals into the execution framework for the compacts

- **Scope:** Battery Materials & Cells; Permanent Magnets & Drive Systems; no other sectors.
- **Reciprocal commitments:** Korea commits demand certainty (procurement recognition and anchor offtake) and market-access predictability for the nodes Australia supplies (rare-earth metals, lithium minerals); Australia commits the same for the nodes Korea supplies (cathode materials & precursors, accumulator parts & separators, non-oriented electrical steel (ngo)). Each commits public finance for the compact nodes, at home, in the partner or with named third-country hosts.
- **Compact portfolio:** Battery Materials & Cells (evidence-led candidate); Permanent Magnets & Drive Systems (evidence-led candidate).
- **Financing lanes:** Korea: KEXIM, K-SURE, KOMIR, KDB; Australia: EFA, NRFC, CEFC, ARENA; demand-side tools (offtake, stockpiles, procurement): MCEE, KOMIR, PPS, NSSC and DFAT, DCCEEW, FIRB; third-country finance: KEXIM, KOMIR, KOICA and none listed.
- **Governance:** one bilateral steering line inside EFA-KEXIM co-financing channel for critical minerals, with the existing sector mechanisms running their workstreams (Comprehensive Strategic Partnership and Critical Minerals Working Group). Projects are selected by a joint committee, not by one side.
- **Milestones:** the 12-month milestones of the deal pipeline (Section 9).
- **KPI dashboard:** per compact node, the coercive share of each country's supply, the bilateral import share, offtake signed (volume), capacity at FID and operating, and public money committed vs private.
- **Review cycle:** every 12 months, with the OSI tables re-run on new trade data.
- **Sunset / exit:** each compact carries the exit conditions in Section 7; the framework expires with the investment vehicle () unless renewed on evidence.

9. 12–36 Month Deal Pipeline

Inference. Ranked lexicographically on separate fields (importance, then additionality, urgency, tractability, execution speed, value-chain interdependency, evidence quality); urgency is not importance. At most two deals per compact.

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

RANK	DEAL / ACTION	COMPACT	INSTRUMENT	RESPONSIBLE INSTITUTIONS	12M MILESTONE	IMPORTANCE	URGENCY	ADDITIONALITY
1	Tariff binding for Korea batteries assembly (li-ion cells & packs, other accumulators) entering Australia	SC-BATTERIES	market access	KOR: MOTIR (MOTIE), K-SURE, KOTRA; AUS: DISR, DFAT	Tariff treatment of the node bound or exempted for the trusted supplier	high	high	high

RANK	DEAL / ACTION	COMPACT	INSTRUMENT	RESPONSIBLE INSTITUTIONS	12M MILESTONE	IMPORTANCE	URGENCY	ADDITIONALITY
2	Long-term offtake by Korea and Australia buyers for non-coercive solar refining (copper wire & rod, polysilicon), sized to underwrite new capacity (candidate hosts: MYS, JPN, DEU)	SC-GRID	anchor offtake	KOR: MCEE, PPS; AUS: DCCEEW	Signed long-term offtake from at least one large buyer	high	high	high
3	Tariff binding for Korea electric vehicles components (accumulator parts & separators, electric motors, ferrite & other magnets) entering Australia	SC-MAGNETS	market access	KOR: MOTIR (MOTIE), K-SURE, KOTRA; AUS: DISR, DFAT	Tariff treatment of the node bound or exempted for the trusted supplier	high	high	medium
4	Tariff binding for Korea wind assembly (electricity meters, overhead conductors, power cables) entering Australia	SC-GRID	market access	KOR: MOTIR (MOTIE), K-SURE, KOTRA; AUS: DISR, DFAT	Tariff treatment of the node bound or exempted for the trusted supplier	high	high	medium
5	Tariff binding for Korea wind towers entering Australia	SC-WIND	market access	KOR: MOTIR (MOTIE), K-SURE, KOTRA; AUS: DISR, DFAT	Tariff treatment of the node bound or exempted for the trusted supplier	high	high	medium
6	Multi-year supply agreements for Korea electric vehicles assembly (electric & plug-in cars, electric buses, trucks & tractors, electric two-wheelers & works trucks) into Australia buyers	SC-EV	strategic imports	KOR: MOTIR (MOTIE), K-SURE, KOTRA; AUS: DISR, DFAT	Named import lanes and volumes agreed between buyers and the supplier's producers	high	high	medium
7	Export-credit and investment-insurance cover for coal refining transactions		export credit	KOR: KEXIM, K-SURE, KOMIR, KDB; AUS: EFA, NRFC, CEFC, ARENA	Cover offered for named transactions	high	low	medium
8	Long-term offtake by Korea buyers for Australia batteries extraction (cobalt ores, lithium minerals, natural graphite), sized to underwrite new capacity (candidate hosts: CHL, ARG, BRA)	SC-BATTERIES	anchor offtake	KOR: MCEE, PPS; AUS: DCCEEW	Signed long-term offtake from at least one large buyer	high	high	low
9	Long-term offtake by Korea and Australia buyers for non-coercive wind refining (copper wire & rod, rare-earth metals, rare-earth oxides & compounds), sized to underwrite new capacity (candidate hosts: THA, USA, MYS)	SC-MAGNETS	anchor offtake	KOR: MCEE, PPS; AUS: DCCEEW	Signed long-term offtake from at least one large buyer	high	high	low

Sources: 1.

10. What Not to Do

- *Recommendation.* **N01** Do not create a new bilateral energy MOU or a free-standing MEA institution: KRAU-M3 already exists (finance and projects but no published list); dock the compacts in it.
- *Recommendation.* **N02** Do not put new public money behind Batteries extraction (cobalt ores, lithium minerals, natural graphite) investment that is already committed (evidence KRAU-14; KRAU-19; KRAU-20); support is additional only beyond it.
- *Recommendation.* **N03** Do not put new public money behind Batteries refining (battery materials (foil, electrolyte), cathode materials & precursors, cobalt intermediates & chemicals) investment that is already committed (evidence KRAU-11; KRAU-12; KRAU-13; KRAU-21); support is additional only beyond it.

- *Recommendation.* **N04** Do not put new public money behind Electric Vehicles extraction (cobalt ores, lithium minerals, natural graphite) investment that is already committed (evidence USAU-09; KRAU-14; KRAU-19; KRAU-20); support is additional only beyond it.
- *Recommendation.* **N05** Do not defend AC generators as a bilateral priority: trusted supply is adequate (AIB LOW INTERVENTION), yet Australia's observed protection or attention is high.
- *Recommendation.* **N06** Do not defend Automatic controllers as a bilateral priority: trusted supply is adequate (AIB LOW INTERVENTION), yet Korea's observed protection or attention is high.

11. Evidence Register

- Every recommendation (IDs J, C, T, I, X, D, N, M) has rows in evidence_register.csv: the OSI metrics behind it (variable, value for each country, year, source, evidence specificity) and any external items, with URL and access date. Recommendations that rest on external research only carry basis = external.[1]
- External evidence vintage: accessed 2026-11-06; 256 items (partial 197, unverified 18, verified 41).[1]

Where the findings are weak or counterintuitive

- Joint bets resting on a node need inherited from its cell (a proxy): Wind towers; Power cables.
- Evidence-led compacts (OSI cannot see the partner's capability in trade data): Battery Materials & Cells; Permanent Magnets & Drive Systems.
- Australia has no listed trade position (under 1% of world supply) at nodes where Korea has material demand: wafers, polysilicon, synthetic graphite (anode-grade), rare-earth oxides & compounds, natural graphite, battery materials (foil, electrolyte), battery recycling feed, electric motors, rare-earth permanent magnets, electric buses, trucks & tractors, pv cover glass, automatic controllers, ac generators, refrigerants, graphite electrodes, semiconductor manufacturing equipment, hydrogen, ammonia & methanol, nuclear special materials, silver powder, direct-reduced iron (dri/hbi). Trade data miss capability that is owned abroad or sold at home, so these nodes cannot show complementarity even where corporate evidence suggests it.
- Low-evidence units among the active states: none.
- Fuel-trade lanes reach tier 1 (coal refining; buyer: both): real energy-security lanes, but outside the electro-industrial compacts and with no market failure beyond the supply concentration the engine measures.

Sources

1. Allied Industrial Base Explorer tables: six-stage strategic position (Competitive Position, supply-chain and deployment need, Strategic Industrial Value, NIPO and IEA PAMS policy priority; scripts/26), the Allied Industrial Base strategic layer (node trade and production shares, allied demand, node classes, compact roles; scripts/31-33), bilateral imports by technology and stage (UN Comtrade 2024; scripts/15) and development-partnership pairs (scripts/93). Underlying sources are listed in the country strategy briefs. Data as of 30 October 2026.
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Appendix — evidence tables and method

The tables below carry the full complementarity map, the node profiles and the trade actions that the main text summarises. Method, decision rules and limitations: docs/partnership_memos.md. Machine-readable outputs: country_profiles.csv, dyad_complementarity.csv, compact_candidates.csv, trade_priorities.csv, investment_priorities.csv, third_country_opportunities.csv, deal_pipeline.csv and evidence_register.csv.

Table A1. Section 2, What Each Country Is Good At — and What It Needs

COMPACT	NODE	STAGE	KOREA SHARE	KOREA NEED	AUSTRALIA SHARE	AUSTRALIA NEED	AIB CLASS	COERCIVE TOP SUPPLIER
SC-BATTERIES	Lithium minerals	Extraction	0.3%	high 1.00	52.7%	low (cell 0.72; not a material buyer)	LOW INTERVENTION	CHN 25%
SC-BATTERIES	Natural graphite	Extraction	2.6%	high 1.00	0.0%	low (cell 0.72; not a material buyer)	BUILD	CHN 82%
SC-BATTERIES	Battery recycling feed	Recovery	0.3%	high 0.94	0.0%	low (cell 0.95; not a material buyer)	LOW INTERVENTION	CHN 8%
SC-BATTERIES	Cobalt intermediates & chemicals	Refining	0.8%	high 0.99	1.6%	low (cell 0.83; not a material buyer)	BUILD	CHN 76%
SC-BATTERIES	Lithium chemicals	Refining	0.7%	high 0.99	2.0%	low (cell 0.83; not a material buyer)	BUILD	CHN 74%
SC-BATTERIES	Nickel intermediates & chemicals	Refining	1.0%	low (cell 0.99; not a material buyer)	3.5%	low (cell 0.83; not a material buyer)	LOW INTERVENTION	CHN 37%
SC-BATTERIES	Cathode materials & precursors	Refining	50.9%	low (cell 0.99; not a material buyer)	0.0%	low (cell 0.83; not a material buyer)	BUILD	CHN 92%
SC-BATTERIES	Accumulator parts & separators	Components	12.7%	low (cell 0.97; not a material buyer)	0.0%	low (cell 0.76; not a material buyer)	BUILD	CHN 38%
SC-BATTERIES	Synthetic graphite (anode-grade)	Refining	4.6%	high 0.99	0.0%	low (cell 0.83; not a material buyer)	BUILD	CHN 94%
SC-BATTERIES	Battery materials (foil, electrolyte)	Refining	9.5%	high 0.99	0.0%	low (cell 0.83; not a material buyer)	BUILD	CHN 16%
SC-BATTERIES	Li-ion cells & packs	Assembly	4.0%	low (cell 0.97; not a material buyer)	0.0%	high 0.85	BUILD	CHN 84%
SC-BATTERIES	Other accumulators	Assembly	1.1%	low (cell 0.83; not a material buyer)	0.2%	high 0.85	BUILD	CHN 28%
SC-EV	Electric & plug-in cars	Assembly	7.0%	low 0.37	0.0%	high 0.70	BUILD	CHN 69%
SC-EV	Electric two-wheelers & works trucks	Assembly	1.1%	low 0.37	0.0%	high 0.70	BUILD	CHN 30%
SC-EV	Electric buses, trucks & tractors	Assembly	0.4%	demand 0.37	0.0%	low (cell 0.70; not a material buyer)	BUILD	CHN 24%
SC-EV	Electric rail vehicles	Assembly	10.6%	low n/a	0.0%	demand n/a	LOW INTERVENTION	CHN 14%
SC-GRID	Copper ores & concentrates	Extraction	0.0%	high 0.97	5.3%	low (cell 0.86; not a material buyer)	LOW INTERVENTION	CHN 13%
SC-GRID	Copper smelting & refining	Refining	1.7%	high 0.96	3.5%	low 0.60	LOW INTERVENTION	CHN 50%
SC-GRID	Grain-oriented electrical steel (GOES)	Refining	5.2%	low (cell 0.96; not a material buyer)	0.0%	low 0.60	BUILD	CHN 41%
SC-GRID	Power cables	Components	8.4%	low 0.64	0.1%	high 0.83	LOW INTERVENTION	CHN 19%
SC-GRID	Electricity meters	Assembly	0.0%	low 0.64	0.1%	high 0.83	BUILD	CHN 34%

COMPACT	NODE	STAGE	KOREA SHARE	KOREA NEED	AUSTRALIA SHARE	AUSTRALIA NEED	AIB CLASS	COERCIVE TOP SUPPLIER
SC-GRID	Overhead conductors	Components	1.2%	low 0.64	0.0%	high 0.83	LOW INTERVENTION	CHN 24%
SC-GRID	Switchgear & protection	Assembly	3.7%	low 0.64	0.5%	low (cell 0.83; not a material buyer)	LOW INTERVENTION	CHN 20%
SC-GRID	Transformers	Assembly	9.8%	low 0.64	0.1%	low (cell 0.83; not a material buyer)	LOW INTERVENTION	CHN 21%
SC-HEAT	Refrigerants	Refining	0.1%	demand n/a	0.0%	not measured n/a	BUILD	CHN 45%
SC-HEAT	Refrigeration & heat-pump compressors	Components	7.0%	low n/a	0.0%	not measured n/a	BUILD	CHN 33%
SC-HEAT	Heat pumps & AC units	Assembly	1.9%	low n/a	0.0%	demand n/a	BUILD	CHN 49%
SC-INDUSTRY	Electrolysers	Assembly	5.7%	low n/a	0.3%	low n/a	BUILD	CHN 32%
SC-INDUSTRY	Hydrogen, ammonia & methanol	Refining	0.2%	demand n/a	0.0%	not measured n/a	LOW INTERVENTION	RUS 3%
SC-INDUSTRY	Furnace parts	Components	2.2%	low n/a	0.0%	not measured n/a	BUILD	CHN 21%
SC-INDUSTRY	Graphite electrodes	Components	0.0%	demand n/a	0.0%	low n/a	COORDINATE	CHN 38%
SC-INDUSTRY	Direct-reduced iron (DRI/HBI)	Refining	0.0%	demand n/a	0.0%	not measured n/a	LOW INTERVENTION	RUS 20%
SC-INDUSTRY	Electric furnaces & heating	Assembly	5.3%	low n/a	0.0%	not measured n/a	BUILD	CHN 27%
SC-MAGNETS	Rare-earth ores (monazite)	Extraction	0.0%	low (cell 1.00; not a material buyer)	5.1%	low (cell 0.72; not a material buyer)	INSUFFICIENT DATA	CHN 59%
SC-MAGNETS	Rare-earth oxides & compounds	Refining	1.0%	high 0.99	0.0%	low (cell 0.86; not a material buyer)	BUILD	CHN 84%
SC-MAGNETS	Rare-earth metals	Refining	0.2%	low (cell 0.99; not a material buyer)	25.1%	low (cell 0.86; not a material buyer)	BUILD	CHN 84%
SC-MAGNETS	Non-oriented electrical steel (NGO)	Refining	13.7%	low (cell 0.96; not a material buyer)	0.0%	low 0.60	BUILD	CHN 26%
SC-MAGNETS	Rare-earth permanent magnets	Components	0.9%	high 0.97	0.0%	low (cell 0.92; not a material buyer)	BUILD	CHN 64%
SC-MAGNETS	Ferrite & other magnets	Components	4.4%	low (cell 0.97; not a material buyer)	0.0%	low (cell 0.92; not a material buyer)	BUILD	CHN 58%
SC-MAGNETS	Electric motors	Assembly	2.6%	high 0.97	0.1%	low (cell 0.76; not a material buyer)	BUILD	CHN 26%
SC-MAGNETS	AC generators	Assembly	0.3%	high 0.80	0.1%	high 0.92	LOW INTERVENTION	CHN 15%
SC-NUCLEAR	Uranium/fuel Extraction: aggregated low-coverage sub-sectors (Spent fuel & nuclear scrap, Uranium ores)	Extraction	0.0%	low (cell 0.96; not a material buyer)	75.0%	low 0.61	INSUFFICIENT DATA	RUS 8%
SC-NUCLEAR	Uranium conversion & enrichment	Refining	0.0%	high 0.98	2.6%	low 0.61	LOW INTERVENTION	RUS 16%
SC-NUCLEAR	Fuel assemblies & zirconium	Components	7.4%	low 0.54	0.0%	low 0.56	LOW INTERVENTION	RUS 28%
SC-NUCLEAR	Nuclear special materials	Refining	0.1%	demand n/a	0.0%	not measured n/a	LOW INTERVENTION	0%

COMPACT	NODE	STAGE	KOREA SHARE	KOREA NEED	AUSTRALIA SHARE	AUSTRALIA NEED	AIB CLASS	COERCIVE TOP SUPPLIER
SC-SEMIS	Gallium, germanium & indium	Refining	18.0%	low n/a	0.0%	not measured n/a	BUILD	CHN 25%
SC-SEMIS	Semiconductor manufacturing equipment	Assembly	5.6%	demand 0.25	0.0%	low (cell 0.81; not a material buyer)	LOW INTERVENTION	CHN 5%
SC-SEMIS	Power semiconductors	Components	3.2%	low (cell 0.70; not a material buyer)	0.0%	low 0.56	BUILD	CHN 32%
SC-SEMIS	Sensors & controller parts	Components	2.4%	low n/a	0.0%	not measured n/a	BUILD	CHN 24%
SC-SEMIS	Integrated circuits	Components	17.1%	low n/a	0.0%	not measured n/a	LOW INTERVENTION	CHN 16%
SC-SEMIS	Power electronics components	Components	2.5%	low (cell 0.97; not a material buyer)	0.0%	low (cell 0.92; not a material buyer)	BUILD	CHN 37%
SC-SEMIS	Automatic controllers	Assembly	2.8%	demand 0.51	0.2%	low 0.65	LOW INTERVENTION	CHN 12%
SC-SOLAR	Polysilicon	Refining	1.2%	high 0.71	0.0%	low (cell 0.81; not a material buyer)	LOW INTERVENTION	CHN 97%
SC-SOLAR	Wafers	Refining	6.4%	demand 0.30	0.0%	low 0.56	BUILD	CHN 98%
SC-SOLAR	Silver powder	Refining	1.6%	demand n/a	0.0%	not measured n/a	LOW INTERVENTION	0%
SC-SOLAR	Solar cells	Components	11.1%	low 0.30	0.0%	low 0.56	BUILD	CHN 92%
SC-SOLAR	PV cover glass	Components	1.9%	demand 0.30	0.0%	low 0.56	BUILD	CHN 61%
SC-WIND	Wind towers	Components	11.7%	low (cell 0.80; not a material buyer)	0.0%	high 0.92	LOW INTERVENTION	CHN 18%
SC-WIND	Wind blades & hubs	Components	1.1%	low (cell 0.80; not a material buyer)	0.1%	high 0.92	BUILD	CHN 71%
SC-WIND	Hydro turbine parts	Components	0.0%	not measured n/a	0.1%	demand n/a	COORDINATE	CHN 35%
SC-WIND	Wind turbine generating sets	Assembly	0.1%	low 0.64	0.0%	high 0.83	BUILD	CHN 74%

Sources: 1.

Table A2. Section 3, Bilateral Complementarity Map

RANK	TIER	TECHNOLOGY	UNIT	STAGE	KOREA POSITION	AUSTRALIA POSITION	DYADIC RELATIONSHIP	STRATEGIC CASE
1	joint bet	Batteries	Batteries assembly (li-ion cells & packs, other accumulators)	Assembly	strong	weak	Korea supplies / Australia needs	geopolitical chokepoint; need 0.83 / 0.85
2	joint bet	Batteries	Batteries components (accumulator parts & separators)	Components	moderate	weak	Korea supplies / Australia needs	strategic manufacturing gap; need 0.73 / 0.72
3	joint bet	Electric Vehicles	Electric Vehicles components (accumulator parts & separators, electric motors, ferrite & other magnets)	Components	strong	weak	Korea supplies / Australia needs	strategic manufacturing gap; need 0.97 / 0.76
4	joint bet	Electric Vehicles	Electric Vehicles assembly (electric & plug-in cars, electric buses, trucks & tractors, electric two-wheelers & works trucks)	Assembly	strong	weak	Korea supplies / Australia needs	strategic manufacturing gap; need 0.37 / 0.70
5	joint bet	Solar	Solar refining (copper wire & rod, polysilicon)	Refining	weak	weak	Shared vulnerability	geopolitical chokepoint; need 0.71 / 0.81
6	joint bet	Wind	Wind components (ac generators, ferrite & other magnets, insulators & bushings)	Components	moderate	weak	Shared vulnerability	strategic manufacturing gap; need 0.80 / 0.92
7	joint bet	Wind & Hydropower Equipment	Wind towers	Components	strong (11.7% of world supply)	weak (0.0% of world supply)	Korea supplies / Australia needs	strategic manufacturing gap; need 0.80 / 0.92
8	joint bet	Wind	Wind assembly (electricity meters, overhead conductors, power cables)	Assembly	moderate	weak	Korea supplies / Australia needs	strategic manufacturing gap; need 0.64 / 0.83

RANK	TIER	TECHNOLOGY	UNIT	STAGE	KOREA POSITION	AUSTRALIA POSITION	DYADIC RELATIONSHIP	STRATEGIC CASE
9	joint bet	Grid Equipment & Conductors	Power cables	Components	strong (8.4% of world supply)	weak (0.1% of world supply)	Korea supplies / Australia needs	strategic manufacturing gap; need 0.64 / 0.83
10	joint bet	Coal	Coal refining	Refining	weak	moderate	Shared vulnerability	strategic manufacturing gap; need 0.93 / 0.72
11	coordinate	Batteries	Batteries refining (battery materials (foil, electrolyte), cathode materials & precursors, cobalt intermediates & chemicals)	Refining	weak	moderate	Shared vulnerability	geopolitical chokepoint; need 0.99 / 0.83
12	coordinate	Electric Vehicles	Electric Vehicles refining (rare-earth metals, rare-earth oxides & compounds)	Refining	weak	weak	Shared vulnerability	geopolitical chokepoint; need 0.99 / 0.86
13	coordinate	Battery Materials & Cells	Cathode materials & precursors	Refining	strong (50.9% of world supply)	weak (0.0% of world supply)	Complementary specialisation	geopolitical chokepoint; need 0.99 / 0.83
14	coordinate	Wind	Wind refining (copper wire & rod, rare-earth metals, rare-earth oxides & compounds)	Refining	weak	weak	Shared vulnerability	geopolitical chokepoint; need 0.75 / 0.84
15	coordinate	Permanent Magnets & Drive Systems	Rare-earth metals	Refining	weak (0.2% of world supply)	strong (25.1% of world supply)	Complementary specialisation	geopolitical chokepoint; need 0.99 / 0.86
16	coordinate	Battery Materials & Cells	Accumulator parts & separators	Components	strong (12.7% of world supply)	weak (0.0% of world supply)	Complementary specialisation	trusted partner capacity constraint; need 0.97 / 0.76
17	coordinate	Batteries	Batteries extraction (cobalt ores, lithium minerals, natural graphite)	Extraction	weak	strong	Capacity-constrained trusted supply	geopolitical chokepoint; need 1.00 / 0.12
18	coordinate	Battery Materials & Cells	Lithium minerals	Extraction	weak (0.3% of world supply)	strong (52.7% of world supply)	Australia supplies / Korea needs	no material intervention; need 1.00 / 0.72
19	coordinate	Electric Vehicles	Electric Vehicles extraction (cobalt ores, lithium minerals, natural graphite)	Extraction	weak	strong	Capacity-constrained trusted supply	geopolitical chokepoint; need 1.00 / 0.72
20	coordinate	Permanent Magnets & Drive Systems	Non-oriented electrical steel (NGO)	Refining	strong (13.7% of world supply)	weak (0.0% of world supply)	Complementary specialisation	trusted partner capacity constraint; need 0.96 / 0.60
21	coordinate	Gas	Gas assembly	Assembly	strong	weak	Complementary specialisation	strategic manufacturing gap; need 0.34 / 0.57
22	coordinate	Coal	Coal components	Components	strong	weak	Complementary specialisation	strategic manufacturing gap; need 0.58 / 0.63
23	coordinate	Coal	Coal assembly	Assembly	strong	weak	Complementary specialisation	strategic manufacturing gap; need 0.60 / 0.50
24	coordinate	Gas	Gas extraction	Extraction	weak	strong	Australia supplies / Korea needs	no material intervention; need 0.96 / 0.28
25	coordinate	Coal	Coal extraction	Extraction	weak	strong	Australia supplies / Korea needs	no material intervention; need 0.94 / 0.16

Sources: 1.

Table A3. Section 4, Trade Agenda

ID	DIRECTION	UNIT	ACTION TYPE	ACTION
T01	KOR -> AUS	Batteries/Assembly	tariff risk	Bind market access for Korea Batteries assembly (li-ion cells & packs, other accumulators) into Australia: Australia's protection priority on this node is in its top third (0.78), which taxes the trusted supplier it needs.
T02	KOR -> AUS	Batteries/Components	tariff risk	Bind market access for Korea Batteries components (accumulator parts & separators) into Australia: Australia's protection priority on this node is in its top third (0.83), which taxes the trusted supplier it needs.
T03	KOR -> AUS	Electric Vehicles/Components	tariff risk	Bind market access for Korea Electric Vehicles components (accumulator parts & separators, electric motors, ferrite & other magnets) into Australia: Australia's protection priority on this node is in its top third (0.99), which taxes the trusted supplier it needs.
T04	KOR -> AUS	Electric Vehicles/Assembly	predictability	Keep Korea exports of Electric Vehicles assembly (electric & plug-in cars, electric buses, trucks & tractors, electric two-wheelers & works trucks) to Australia tariff-free and predictable; the buyer's need is met by the trusted supplier.
T05	both	Solar/Refining	anti circumvention	Common origin and anti-circumvention rules for Solar refining (copper wire & rod, polysilicon): both import heavily from CHN, so a measure one side takes alone leaks through the other.

ID	DIRECTION	UNIT	ACTION TYPE	ACTION
T06	both	Wind/Components	common standards	Common specifications for Wind components (ac generators, ferrite & other magnets, insulators & bushings) so a joint supplier base can serve both markets.
T07	KOR -> AUS	wind-towers	tariff risk	Bind market access for Korea Wind towers into Australia: Australia's protection priority on this node is in its top third (0.68), which taxes the trusted supplier it needs.
T08	KOR -> AUS	Wind/Assembly	tariff risk	Bind market access for Korea Wind assembly (electricity meters, overhead conductors, power cables) into Australia: Australia's protection priority on this node is in its top third (0.96), which taxes the trusted supplier it needs.
T09	KOR -> AUS	power-cables	tariff risk	Bind market access for Korea Power cables into Australia: Australia's protection priority on this node is in its top third (0.88), which taxes the trusted supplier it needs.
T10	both	Coal/Refining	common standards	Common specifications for Coal refining so a joint supplier base can serve both markets.
T11	both	Batteries/Refining	anti circumvention	Common origin and anti-circumvention rules for Batteries refining (battery materials (foil, electrolyte), cathode materials & precursors, cobalt intermediates & chemicals): both import heavily from CHN, so a measure one side takes alone leaks through the other.
T12	both	Electric Vehicles/Refining	anti circumvention	Common origin and anti-circumvention rules for Electric Vehicles refining (rare-earth metals, rare-earth oxides & compounds): both import heavily from CHN, so a measure one side takes alone leaks through the other.
T13	open	nuclear-special-materials	open sourcing	Leave Nuclear special materials open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.
T14	open	battery-recycling-feed	open sourcing	Leave Battery recycling feed open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.
T15	open	semiconductor-manufacturing-equipment	open sourcing	Leave Semiconductor manufacturing equipment open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.
T16	open	copper-ores-and-concentrates	open sourcing	Leave Copper ores & concentrates open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.
T17	open	silver-powder	open sourcing	Leave Silver powder open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.
T18	open	uranium-conversion-and-enrichment	open sourcing	Leave Uranium conversion & enrichment open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.

Sources: 1.

Rule-generated from the data (R/partnerships) with external facts from a dated, curated evidence layer; not yet checked by an independent verifier. Items marked partial or unverified must be confirmed before citation.
Allied Industrial Base Explorer, Partnership memo: Korea–Australia Strategic Energy & Industrial Partnership